AP Statistics
Unit 7

Name:_____

7A Confidence Intervals for a Population Mean or Mean Difference

1. City council members want to estimate how many pounds of trash households in their community produce per week. To determine an estimate for the standard deviation of the weight of trash produced a small random sample of households was selected and their trash was weighed on garbage day. This produced an estimated standard deviation of 36 pounds.

   a. How many households would need to be surveyed to estimate the population mean at the 95% confidence level with a margin of error of at most 3 pounds?

   b. After solving part (a), the city council realizes that it would be too much work to weigh the garbage of that many households. They give up on their hopes of estimating the true mean weight of trash produced within 3 pounds and select a random sample of 15 households and weigh the trash for each of these households on garbage day. Here are the results:

   114.8 74.3 80.1 41.5 99.1 31.0 93.1 118.9 26.5 33.1 88.3 46.1 119.7 46.4 19.8

   Calculate and interpret a 95% confidence interval for the mean weight of trash for all households in this community. Use the 4-Step Process!

State the Problem:

Parameter:

Confidence level:

Plan:

Procedure:

Conditions:

Calculate:

General Formula:

Specific Formula:

Work/Solution:

Conclude:

2. An educational researcher wanted to know if there was a difference in student learning between lectures paired with in-class activities compared to traditional lecture-only teaching methods. Ten students matched on GPA, year in school, and academic major were randomly selected from the student population at a local university. One student from each pair was randomly assigned to either the "Lecture Only" class or "Lecture & Group Activities" class. At the end of the semester, students' final exam scores were recorded and teaching methods were compared.

Group #	1	2	3	4	5	6	7	8	9	10
Lecture	82	73	77	71	80	91	87	90	72	83
Lecture & Activity	88	72	84	74	91	93	87	89	80	87

Estimate the difference in final exam score between traditional lecture teaching method and lecture paired with in-class activities teaching method.

State the Problem:

Parameter:

Confidence level:

Plan:

Procedure:

Conditions:

Calculate:

Specific Formula:

Work/Solution:

Conclude:

3. The Deerfield baseball team hosts clinics through DYBA advertising that their clinics will improve the ability of Mustang players to throw strikes. To test this claim, they take 20 random Mustang players signed up for the spring season and have them throw 50 pitches each and record the number of strikes. After they participate in the clinic, they repeat the test. The computer output shows the resulting data summary.

a. Create a 95% confidence interval to estimate the difference in strikes.
b. Using your answer from part (a) and *without* running a hypothesis test, is the convincing evidence that the clinic is effective? Explain.

t-inference

	N	Mean	Std Dev
Before	20	33.05	2.438
After	20	34.95	2.837
Diff	20	–1.90	2.606

7B Significance Tests for a Population Mean or Mean Difference

1. It is recommended that teenagers get 8 hours of sleep a night. Mrs. Bassler believes that her AP Statistics students are getting less than the recommended 8 hours of sleep per night. To test her belief, she takes a random sample of 10 students from her 3 sections of AP Statistics and records the number of hours of sleep for student (see below). Do these data provide convincing evidence that the AP Statistics students get less than 8 hours of sleep per night? Use a significance level of $\alpha = 0.05$.

Hours of sleep last night: 6.5 4.75 5 8 7 7.25 4.5 5.5 7 6

State the Problem:

Parameter:

Statistic:

Hypotheses:

Significance level:

Plan:

Procedure:

Conditions:

Calculate:

(Formula, Work, Picture, Test Statistic, P-value)

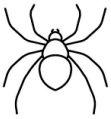
Conclude:

2. In 1974, the Bellevue-Stratford Hotel in Philadelphia was the scene of an outbreak of what later became known as legionnaires' disease. The cause of the disease was finally discovered to be bacteria that thrived in the air-conditioning units of the hotel. Owners of a hotel down the street, hearing about the Bellevue-Stratford, replaced their air-conditioning system. The following data are the bacteria counts in the air of eight rooms, before and after a new air-conditioning system was installed (measured in colonies per cubic foot of air).

   Is there evidence that the new system succeeded in lowering the bacterial count?

Room #	Before	After
121	11.8	10.1
163	8.2	7.2
125	7.1	3.8
264	14	12
233	10.8	8.3
218	10.1	10.5
324	14.6	12.1
325	14	13.7

3. In many southern U.S. states, porch ceilings are painted in a light blue color. Some people believe that the color tricks spiders, wasps, and other insects into thinking that the ceiling is the sky and that this reduces the number of insects found on the porches.



Debbie wonders if this is really true and decides to investigate. She enlists the help of 35 homeowners with porches in her South Carolina county. On each of these homes she paints half of the porch ceiling white and the other half of the porch ceiling blue. The half that is painted blue is determined by tossing a coin. Debbie also places an insect trap on both sides of each porch.

After a period of two weeks, Debbie counts the number of insects caught in each trap and determines the average number of insects caught near each color of paint. Her results are shown in the table to the right.

	Sample mean	Sample standard deviation
Blue Paint	86.4	4.9
White Paint	87.7	5.2
Diff (B – W)	–1.3	7.1

Is there evidence to prove that blue ceilings reduce the number of insects found on porches?

Comparing 1-Sample t-Intervals & 1-Sample t-Tests

It is commonly thought that "normal" body temperature is 98.6 degrees Fahrenheit. A statistics class wants to find out if this is really true or if "normal" body temperature is actually different from 98.6°F. They take a random sample of 30 students at their high school and record their body temperatures. They get a sample mean of 97.9°F with a standard deviation of 1.6°F.

a. Suppose all conditions for inference have been met. Use your calculator to find the P-value for a 1-Sample t-Test.

b. Decide if you would reject the null hypothesis that the true mean body temperature is 98.6°F at the different significance levels listed below.

Reject H_0 at $\alpha = 0.10$? _____ Reject H_0 at $\alpha = 0.05$? _____ Reject H_0 at $\alpha = 0.01$? _____

c. Use your calculator to find a confidence interval to estimate the true mean body temperature for the given significance levels. Then decide if each interval would cause you to reject the null hypothesis that the true mean body temperature is 98.6°F.

90% confidence interval: _____ Reject H_0? _____

95% confidence interval: _____ Reject H_0? _____

99% confidence interval: _____ Reject H_0? _____

d. What connection do you notice between your answers to parts b and c?

*Note: this is only true when the significance test is two-sided.

Population Mean or Mean Difference Inference Practice

1. According to the AAA Foundation for Traffic Safety's American Driving Survey, U.S. drivers spend, on average, 51 minutes behind the wheel each day. A researcher believes this is an overstatement. To investigate, a random sample of 75 drivers were selected. The study revealed that the mean time behind the wheel for the sample of 75 drivers was 46.4 minutes with a standard deviation of 18.8 minutes.

   a. Is there convincing evidence that the mean time behind the wheel for all U.S. drivers is less than 51 minutes? Use $\alpha = 0.01$.

   b. Interpret the p-value from part a.

2. A statistics class wanted to estimate the average weight of an Oreo cookie to determine if the average weight was less than advertised. She selected a random sample of 30 cookies and found the weight of each cookie (in grams). The mean weight was $\bar{x}$ = 11.1921 grams with a standard deviation of s_x = 0.0817 grams.

   a. Create a 95% confidence interval to estimate the true mean weight of an Oreo.

   b. According to Nabisco, an Oreo weighs 11.3 grams. Does your confidence interval in part a provide convincing evidence that the true mean weight is less than 11.3 grams? Justify your solution.

3. Many people think it's faster to order at the drive-thru than to order inside at fast-food restaurants. Two student researchers used a random number generator to select 10 times over a 2-week period to visit a local Dunkin' restaurant. At each of these times, one student ordered an iced coffee at the drive-thru and the other ordered an iced coffee at the counter inside. A coin flip determined who went inside and who went to the drive-thru. The table shows the times, in seconds, that it took for each student o receive the iced coffee after placing the order.

Visit	1	2	3	4	5	6	7	8	9	10
Inside Time	62	63	325	105	135	55	92	75	203	100
Drive-Thru Time	55	50	321	110	124	54	90	68	200	103

Do these data provide convincing evidence at the α = 0.05 level that it is faster to order at the drive-thru than to order inside, on average, at this Dunkin' restaurant?

4. Breast feeding mothers secrete calcium into their milk. Some of the calcium may come from their bones, so mothers may lose bone mineral. Researchers measured the percent change in bone mineral content (BMC) of the spines of 47 randomly selected mothers during 3 months of breast-feeding. The mean change in BMC was –3.587% and the standard deviation was 2.506%.

   a. Construct and interpret a 99% confidence interval to estimate the mean percent change in BMC in the population of breast-feeding mothers.

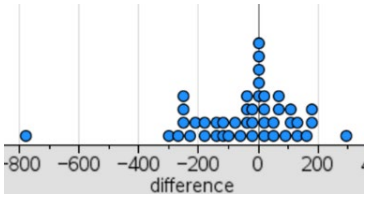
   b. Based on your interval from part (a), do these data give convincing evidence that breastfeeding causes nursing mothers to lose bone mineral, on average? Explain.

5. Hoping to lure more shoppers downtown, a city builds a new public parking garage in the central business district. The city plans to pay for the structure through parking fees. During a two-month period (44 weekdays), the daily fees collected averaged $126, with a standard deviation of $15.

   a. Create a 90% confidence interval to estimate the mean daily income this parking garage will generate.

   b. Using your answer from part (a) and *without* running a hypothesis test, is there significant evidence to suggest the parking garage will average below $130?

6. Does Friday the 13th have an effect on people's shopping behavior? Researchers collected data on the number of shoppers at a random sample of 45 grocery stores on Friday the 6th and Friday the 13th in the same month. A dotplot of the difference (6th – 13th) in the number of shoppers at each store on these 2 days is shown here. The mean difference is -46.5 and the standard deviation of the differences is 178.



Do these data provide convincing evidence at the 10% significance level that the number of shoppers at grocery stores on these 2 days differs, on average?

7. The upper limit for sodium content on ‘reduced sodium’ hot dogs is 325 mg. A nutrition lab tested 40 hot dogs to see if their mean sodium content was less that the limit. Their results found a mean of 322 grams and a standard deviation of 36 grams.

   a. Does this provide significant evidence that the mean sodium content in ‘reduced sodium’ hot dogs is less than 325 grams?

   b. If the nutrition lab made an error in their conclusion, what type of error did they make? What would be the ramifications of this error? Explain.

8. A sleep researcher is curious about if a mindfulness meditation program improves sleep duration. She selects a random sample of 10 adults and records the average number of hours they sleep per night before starting the program. After four weeks of daily meditation, she records their new average sleep duration shown in the table.

Subject	1	2	3	4	5	6	7	8	9	10
Before Meditation	6.2	5.5	7.1	6.8	5.9	6.3	7.2	5.8	6.0	6.4
After Meditation	7.0	6.3	7.8	7.6	6.6	7.0	7.9	7.7	6.8	7.2

Construct a 95% confidence interval to estimate the mean difference in sleep duration for adults using this mediation program.

7C Confidence Intervals for a Difference in Population Means Notes

Purpose of a 2-Sample t-Interval

When ***estimating*** the difference in means from two populations, we use a **2-sample t-interval for the difference in population means**.

Conditions

To estimate a difference in population means, we first need to check the following conditions:

- **Normal**?
  To check that the sampling distribution of the difference in means is approximately normal, **one** of the following conditions must be met for **each** sample (not necessarily the same condition):
  - The sample comes from an approximately normal population.
  - The sample size is greater than 30.
  - The shape of the distribution of the sample should be free from extreme skewness or outliers. (Check this condition with a dotplot/boxplot when the sample size is small.)

- **Random**?
  Study: The data should come from two independent random samples
  Experiment: There should be random assignment

- **Independent**?
  Study: When sampling without replacement, the samples should be less than or equal to 10% of their respective populations (10% condition).
  Experiment: There should be random assignment.

- **Independent Populations**?
  Study: The two populations from which the samples were taken should be independent of each other.
  Experiment: There should be random assignment.

Confidence Interval Formula

You will either be given the necessary information for the confidence interval formula or you will need to enter the data into a Lists & Spreadsheets page your calculator to find that information (see next page).

General Formula: $statistic \pm (critical\ value)(standard\ error\ of\ statistic)$
(Doesn't need to be shown.)

Specific Formula: $CI = (\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}$

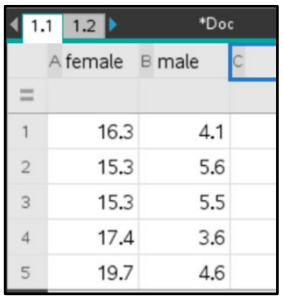
Degrees of Freedom: This is needed to find the critical value, t^*. The formula to find the degrees of freedom for a **2-sample** t-interval or t-test is rather complicated (see below), and is NOT on your formula sheet.

*Always use your calculator to find the degrees of freedom before plugging the numbers into the formula.

$$\mathrm{df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$$

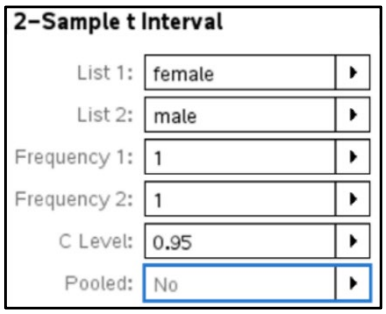
Calculator Instructions

When finding a **confidence interval** for a **difference in means** on the TI-Nspire, you must do the following:



	A female	B male	C
1	16.3	4.1	
2	15.3	5.6	
3	15.3	5.5	
4	17.4	3.6	
5	19.7	4.6	

- If given 2 data sets, enter them in separate lists on a Lists & Spreadsheets page. *(The example contains the data set about spiders from the AP Classroom videos.)*
- Open a Calculator page.
  - Press MENU.
  - Select **6:** Statistics, **6:** Confidence Intervals, **4:** 2-Sample t Interval
  - Since we have data entered into the Lists & Spreadsheets page, select "Data". Otherwise, select "Stats".
  - Enter the list names, the confidence level, and make sure that you select "**No**" when asked to pool the data.
    **Be sure that the variables you use for List 1 and List 2 correspond to the order you used when you defined your parameters.*



- The output gives the confidence interval and all of the statistics needed for the 2-Sample t-interval formula.
  ***Take note of the degrees of freedom. Using the calculator is the only way to find the correct degrees of freedom, which is needed to find the critical value, t*.*

	"Title"	"2-Sample t Interval"
Lower bound of the CI	"CLower"	7.95629694129
Upper bound of the CI	"CUpper"	12.1294173444
Sample mean of the difference (List 1 – List 2)	"x̄Diff"	10.0428571429
Margin of error	"ME"	2.08656020157
Degrees of freedom	"df"	14.7328456476
Sample mean of List 1	"x̄1"	14.15
Sample mean of List 2	"x̄2"	4.10714285714
Standard deviation of List 1	"sx1"	3.54048019342
Standard deviation of List 2	"sx2"	0.916065403875
Sample size of List 1	"n1"	14.
Sample size of list 2	"n2"	14.

7C Confidence Intervals for a Difference in Population Means Practice

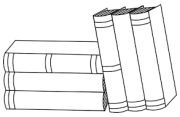
1. Katie uses a ride share company to get to her friend's house. She was wondering what the mean difference would be in wait times (in minutes) for Uber vs. Lyft. Twenty-five randomly selected times are accessed during the day for each company. Graphical displays indicate that the assumption for normality is not unreasonable. Below is the computer output of the data.

	N	Mean	StDev
Uber	25	8.3	1.1
Lyft	25	8.9	1.3
Diff	25	–0.6	1.7

a. Estimate the difference in mean wait times for Uber vs. Lyft.

b. Using your results from part (a), is there a significant difference in wait times between Uber and Lyft?

2. The most recent American Time Use Survey, conducted by the Bureau of Labor Statistics, found that many Americans barely spend any time reading for fun. People ages 15 to 19 average only 7.8 minutes of leisurely reading per day with a standard deviation of 5.4 minutes. However, people ages 75 and over read for an average of 43.8 minutes per day with a standard deviation of 35.5 minutes. These results were based on random samples of 975 people ages 15 to 19 and 1050 people ages 75 and over.



a. Construct and interpret a 95% confidence interval for the difference in mean amount of time (minutes) that people aged 15 to 19 and people ages 75 and over read per day.

b. Interpret the confidence level in part a.

7D Significant Tests for a Difference in Population Means

1. In a certain state, officials at the Department of Education wanted to investigate whether seniors in high school are absent more often than other high school students. At the end of the school year, a random sample of 50 seniors from the state was selected and their attendance records were analyzed. The mean number of days these seniors were absent from school was 6.2 with a standard deviation of 2.1.

   Attendance records were also analyzed for a second, independent random sample of 50 high school students in the state who were not seniors. The mean number of days these students were absent was 5.7 days with a standard deviation of 1.6. No outliers were found in either sample.

   a. Does the data provide convincing evidence at the $\alpha = 0.10$ level that seniors in high school have more absences on average than other high school students?

   b. Interpret the p-value.

2. Can balloons hold more air or more water before bursting? A student purchased a large bag of 12-inch balloons. They randomly selected 10 balloons from the bag. Half were randomly assigned to be filled with air until bursting, and the other half were to be filled with water until bursting. They used devices to measure the among of air and water that was dispensed until the balloons burst. The student's data was recorded in the table shown.

Does this provide convincing evidence that air-filled balloons can attain a greater volume than water-filled balloons?

Air (ft^3)	Water (ft^3)
0.52	0.44
0.58	0.41
0.50	0.45
0.55	0.46
0.61	0.38

3. According to a Stanford Business article, Americans may eat fewer calories at restaurants if the calories of the food items are labeled on the menu. To investigate this, researchers compared Starbucks receipts from locations where the menus were labeled to receipts from stores where the menus were not labeled. A random sample of 30 receipts from stores with the menus labeled had an average number of calories of 225 calories with a standard deviation of 100 calories. A random sample of 40 receipts from stores without menus labeled showed an average of 265 calories per receipt with a standard deviation of 75 calories.

Does this provide convincing evidence that the average calories per receipt at Starbucks with a labeled menu is less than at a Starbucks without labeled menus?

4. How quickly do synthetic fabrics such as polyester decay in landfills? A researcher buried polyester strips in the soil for different lengths of time, then dug up the strips and measured the force required to break them. Breaking strength is easy to measure and is a good indicator of decay. Lower strength means the fabric has decayed more. For one part of the study, the researcher buried 10 strips of polyester fabric in well-drained soil in the summer. The strips were randomly assigned to two groups: 5 of them were buried for 2 weeks and the other 5 were buried for 16 weeks. The table below gives the breaking strength in pounds.

Group 1 (2 weeks)	118	126	126	120	129
Group 2 (16 weeks)	124	98	110	140	110

Do the data give convincing evidence that polyester decays more in 16 weeks than in 2 weeks, on average?